

Legitron: Evaluating Training and Retrieval Strategies for Large Language Models in International Humanitarian Law

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Abstract—General-purpose Large Language Models are often restricted in the legal domain due to liability and reliability risks. Contributing to the development of Legitron, a specialized model for International Humanitarian Law (IHL), we conduct a comparative analysis of various base models and fine-tuning strategies on a targeted IHL benchmark. By evaluating performance within this structured legal framework, we isolate effective methodologies and eliminate suboptimal approaches. This work provides critical empirical insights to guide the future architectural design and training pipelines of the Legitron project.

I. INTRODUCTION

Large Language Models (LLMs) are now widely used across many domains, but their application to legal advice remains restricted due to the risks of providing licensed, situation-specific guidance. For instance, OpenAI explicitly prohibits the provision of tailored legal or medical advice without the involvement of a licensed professional [1]. These restrictions reflect concerns related to liability, misuse, and the potential harm caused by incorrect legal recommendations.

In contrast, certain legal sub-domains lend themselves more readily to structured computational approaches. International Humanitarian Law (IHL) is particularly suitable due to its well-defined normative framework, notably the 161 rules of Customary IHL identified and codified by the ICRC [2]. These rules complement treaty law and provide a consistent, binding foundation that reduces interpretative ambiguity.

This work serves as a preliminary study for Legitron, a project at the Light Lab at EPFL aimed at developing a domain-specific LLM for IHL. Rather than deploying a full legal assistant, we focus on evaluating base models and training strategies on a targeted benchmark. Our results identify both effective and ineffective design choices, providing empirical guidance for future large-scale development of the Legitron system.

II. MODELS AND METHODS

A. Base Model

Legitron is built on an open-weight large language model selected from several publicly available candidates on Hugging Face, including Qwen [3] (Qwen2.5-7B-Instruct), Mistral [4] (Mistral-7B-Instruct-v0.3), and LLaMA [5] (Llama-

3.1-8B-Instruct). These models were evaluated for their reasoning capabilities, instruction-following behavior, and baseline performance on legal-domain text.

The selection criteria focused on consistency, controllability, and the ability to handle normative legal language without relying on explicit chain-of-thought prompting. The final base model was chosen based on preliminary evaluations on International Humanitarian Law benchmark.

B. Pre-training and Instruction Fine-tuning

Starting from the selected base model, we adapt it to legal language through continual pre-training on a corpus of legal texts provided by the Light Lab through their CSCS cluster [6]. This stage follows a standard next-token prediction objective and does not retrain the model from scratch, but instead injects legal-specific vocabulary, phrasing, and stylistic nuances.

Instruction fine-tuning (IFT) is then applied to teach the model to follow user instructions and interact in a conversational setting. This is achieved by training on data formatted with a chat template, enabling the model to handle structured prompts and user queries effectively.

In addition, we constructed a multiple-choice question (MCQ) dataset derived from the original benchmark to improve performance on MCQ-style questions, which are common in both academic evaluations and our experimental setting.

1) *Continual Pre-Training*: The primary objective of the initial phase was to adapt the base model to the legal domain, enriching its internal knowledge representation. Our initial training experiments utilized the pre-existing Legitron mixture datasets; however, evaluating the models revealed a significant degradation in performance compared to base models, suggesting that the training was actively hindering the model’s capabilities. Upon closer inspection, we discovered that these datasets lacked sufficient specific legal content to effectively ground the model in the domain. Consequently, we pivoted to a Continual Pre-Training (CPT) approach using a manually curated corpus of legal texts provided by our supervisor, denoted as *charlotte_scrape* (see datasets samples comparison in Appendix C). While this specialized pre-training restored performance to levels nearly equivalent to base models, it did not surpass it.

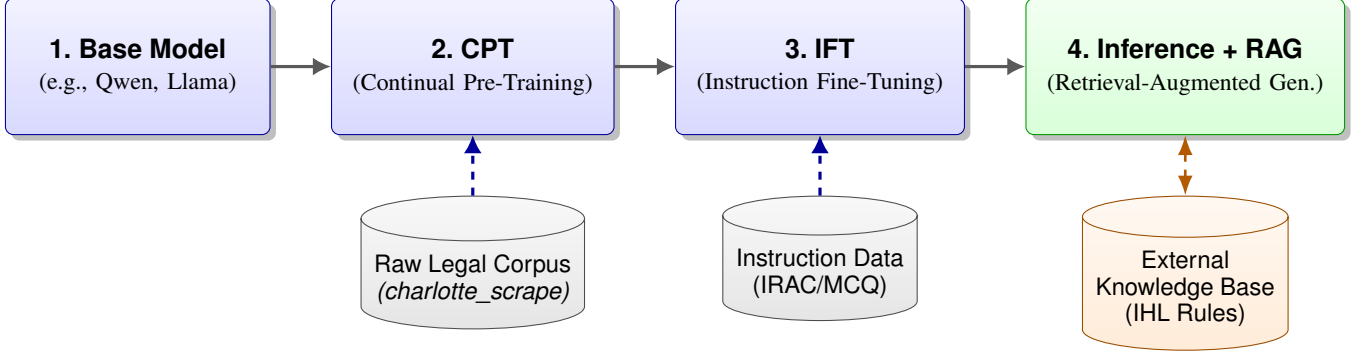


Figure 1. Overview of the experimental methodology pipeline. The process transitions from a generic base model through two stages of training (CPT and IFT) before deploying a RAG mechanism during the final inference phase to ground answers in authoritative IHL rules.

2) *Instruction Fine-Tuning (IFT)*: Following the Continual Pre-Training phase, the second objective was to transition the model from a generalist predictor to a specialized legal reasoner capable of navigating International Humanitarian Law (IHL). To achieve this, we implemented an Instruction Fine-Tuning (IFT) strategy designed to align the model with specific user intents and legal constraints.

Theoretical Background: IFT is a critical post-training process that conditions a Large Language Model (LLM) to respond to directives rather than simply predicting the next token. As demonstrated in foundational work like Instruct-GPT [7], training on explicit input-output pairs significantly reduces hallucinations and improves adherence to complex constraints.

Initial Configuration and Synthetic Data Generation: Given the limitations of the initial laboratory-provided instruction mixture (which lacked domain specificity), we developed a high-throughput synthetic data generation pipeline on the CSCS cluster using vLLM [8]. Leveraging Qwen2.5-32B-Instruct as a “teacher model”, we processed the raw legal texts from the *charlotte_scrape* corpus. The pipeline generated the specialized dataset by prompting a teacher model to produce challenging IHL scenarios within a strict IRAC-structured JSON schema (see Appendix C for a data sample).

Performance Analysis and Strategic Pivot: Despite the successful generation of IRAC-structured data, the subsequent IFT phase did not yield the expected improvements on our evaluation benchmarks. Post-training analysis revealed a critical misalignment between our training objective and the evaluation criteria. While the model improved its ability to generate free-form legal arguments, it failed to transfer this reasoning to the specific constraints of Multiple Choice Questions (MCQ). Consequently, we concluded that the next iteration of the synthetic dataset must explicitly incorporate MCQ-style examples to ensure the model learns to converge its reasoning into a specific option selection, and this IFT configuration was therefore not retained in the final system.

C. Reasoning

We explored several approaches to explicitly enhance reasoning through prompting and iterative interactions with the LLM. In particular, we experimented with a Graph-of-Thoughts framework [9], but this approach proved costly to implement, without yielding statistically significant gains. We also evaluated structured Chain-of-Thought reasoning using the IRAC (Issue, Rule, Application, Conclusion) framework; however, the results were inconclusive and did not justify the added complexity.

Ultimately, we observed that modern base models such as Qwen already exhibit strong built-in reasoning capabilities. As a result, we restricted our approach to a simple Zero-Shot Chain-of-Thought trigger “think quickly step by step” (see prompt example in Appendix A-A, allowing us to prioritize retrieval, data quality, and domain adaptation.

D. Retrieval-Augmented Generation (RAG)

Customary IHL has been systematically identified and codified by the International Committee of the Red Cross (ICRC), which recognizes 161 rules derived from extensive analyses of State practice. These rules reflect binding customary norms applicable in both international and non-international armed conflicts. The authoritative formulation of these rules is publicly available through the ICRC Customary IHL Database [2]. Since these rules constitute the normative foundation upon which legal reasoning should be based, our model should have access to these rules and be able to retrieve them exactly.

Our RAG pipeline follows a straightforward yet effective design. Each of the 161 customary IHL rules is stored in a json file containing the normative rule text and, when available, its accompanying interpretation. All rule texts are embedded using the sentence embedding model BAAI/bge-large-en, producing dense vector representations that capture their semantic content.

For each evaluation question, a query embedding is generated from the question text using the same embedding

model. Cosine similarity is then computed between the query embedding and all rule embeddings in the knowledge base, and the rules are ranked according to their similarity scores. We systematically evaluated different retrieval configurations, including varying the number of retrieved rules (k) and applying similarity thresholds, in order to analyze their impact on retrieval quality.

E. Prompting

Prompt design was found to have a significant impact on model performance. Small modifications in wording could lead to substantial changes in answer quality, and in some cases even a single word affected the outcome. This sensitivity was particularly pronounced for multiple-choice questions, where the structure and constraints of the prompt strongly influenced the model’s behavior.

Prompting also plays a central role in the integration of the RAG mechanism. The retrieved legal rules are injected directly into the prompt, and it is therefore essential to clearly indicate to the model that these rules originate from an authoritative external database. The prompt explicitly instructs the model to base its reasoning and answers on the provided rules (see Appendix A-B), rather than relying on its internal assumptions or general legal knowledge. This explicit grounding was necessary to ensure consistent and source-aware responses.

F. Evaluation Benchmark

The evaluation of *Legitron* is conducted using a benchmark derived from authoritative sources in International Humanitarian Law, provided by the International Committee of the Red Cross (ICRC).

The dataset aggregates 471 legal MCQ originating from multiple ICRC challenge reports published between 2003 and 2024, as well as directly from the corpus of the 161 Customary IHL rules (see Appendix D).

All benchmark questions are formulated as multiple-choice questions with four possible options, where at least one option is correct. Model performance is evaluated using a strict accuracy metric: a score of 1 is assigned only if the predicted answer exactly matches the full set of correct options, while partially correct or incorrect answers receive a score of 0.

G. Experimental Methods

We explored an ensemble-based approach (*AI council*) in which three base models (Qwen, Mistral, and LLaMA) answered each question independently and their outputs were aggregated. This did not improve performance, as the models consistently failed on the same questions, leading to correlated errors and results worse than the best single model (Qwen). Consequently, this strategy was not retained in the final system.

H. Implementation Details

The project infrastructure is built on PyTorch, utilizing the Hugging Face ecosystem (Transformers, SentenceTransformers) for model architecture and embedding generation. For the training pipeline (CPT and IFT), we rely on Axolotl as the primary orchestration framework, which integrates DeepSpeed, Accelerate, and PEFT (with bitsandbytes). Experiment tracking and visualization are managed via Weights & Biases (WandB). The high-throughput synthetic data generation pipeline leverages vLLM for optimized inference speed. Finally, Scikit-learn and NumPy handle downstream metric calculation and vector operations, with tqdm providing real-time progress monitoring during data processing.

All training, inference, and evaluation experiments were conducted on the CSCS high-performance computing infrastructure [10] [6].

III. RESULTS

Table I
PERFORMANCE COMPARISON OF BASE MODELS

Model	Accuracy
Qwen2.5-7B-Instruct	70.28%
Llama-3.1-8B-Instruct	66.88%
Mistral-7B-Instruct-v0.3	65.18%

Base Model: We established a performance baseline by evaluating the three candidate models on the IHL benchmark. As shown in Table I, Qwen outperformed the other models with an accuracy of 70.28%.

Subsequent attempts at Continual Pre-Training (CPT) and Instruction Fine-Tuning (IFT) did not yield conclusive improvements, as discussed earlier. Despite significant effort invested in these approaches, their impact on benchmark performance remained limited. As a result, we decided to discontinue further exploration of CPT and IFT and redirect our focus toward retrieval-based methods.

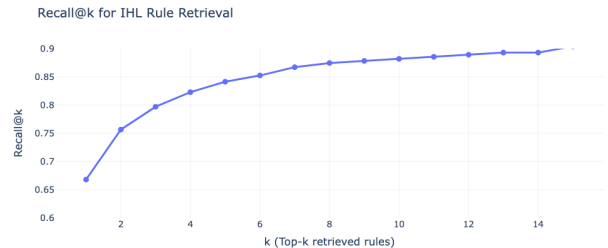


Figure 2. Recall@ k performance of the retrieval component for Customary IHL rule identification based on the benchmark. The curve reports the proportion of questions for which the gold rule appears among the top- k retrieved rules.

RAG: Figure 2 reports the Recall@ k of the retrieval module, defined as the proportion of questions for which

the gold-standard rule appears among the top- k retrieved rules. Recall increases with k and plateaus around $k = 7$. The correct rule is ranked first in 66% of cases, with an average rank of 1.95.

In addition to Recall@ k , we analyze the distribution of cosine similarity scores for the gold-standard rules, shown in Figure 3. The retrieval parameters (k and similarity threshold) were initially selected to $k = 3$ and the similarity threshold to 0.82 using a simple heuristic rather than systematic optimization. Subsequent experiments over k values (1 to 10) showed that these parameters have almost no influence on Qwen’s performance and only a moderate effect on LLaMA. These results suggest that fine-tuning retrieval hyperparameters is not critical in our setting.

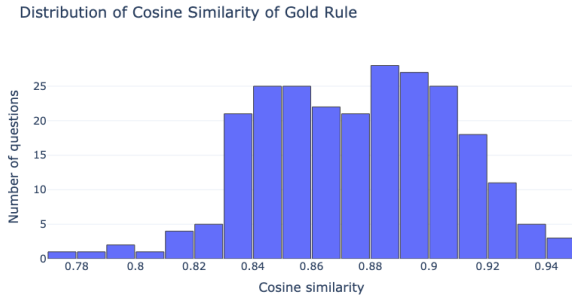


Figure 3. Distribution of the similarity of the golden rule with the question.

Base model with RAG: We evaluated the effectiveness of the base model with the RAG component by measuring the accuracy on the questions (MCQ accuracy) as well as its ability to retrieve the correct IHL rule associated with each benchmark question (Source Accuracy).

Table II
PERFORMANCE COMPARISON USING RAG :
MCQ VS. SOURCE ATTRIBUTION ACCURACY

Model	MCQ Accuracy	Source Accuracy
Qwen2.5-7B-Instruct	71.55%	25.69% (46.18%)
Llama-3.1-8B-Instruct	65.61%	34.39% (61.82%)
Mistral-7B-Instruct-v0.3	58.81%	27.18% (48.86%)

Note: Values in parentheses denote accuracy calculated on the subset of questions citing Customary IHL Rules only ($N = 262$). Questions requiring ICRC reports as sources are excluded from this specific metric, as these documents are not present in the RAG knowledge base.

As detailed in Table II, Qwen demonstrates superior legal reasoning capabilities, achieving the highest MCQ accuracy of **71.55%**. Conversely, Llama excels in grounding, significantly outperforming the other models with a source attribution accuracy of **61.82%** on the subset of customary rules. This divergence highlights a critical trade-off: the model most capable of logical deduction (Qwen) is not necessarily the most reliable at retrieving and citing the correct legal authority. Qwen frequently struggles to comply with

strict prompting requirements, most notably the obligation to output a specific legal rule, while LLaMA consistently provides explicit source citations.

To isolate the impact of retrieval errors, we conducted an "Oracle RAG"[11] experiment by manually injecting the ground-truth source into the prompt. The results showed only marginal improvements: Qwen rose from **70.28% (No RAG)** to **72.82% (Oracle)**.

This narrow gap of around 2.54% indicates that the primary bottleneck is not retrieval. Even in a perfect legal context, the models misapply the rules. However, despite the limited accuracy boost, the RAG remains essential for the project. It ensures that answers are grounded in verifiable IHL rules, which is a critical requirement for transparency in legal AI, regardless of the raw accuracy score.

IV. SUMMARY

This study established a foundational benchmark for specialized International Humanitarian Law (IHL) models. Through a comparative analysis of base models and retrieval strategies, we identified a critical trade-off: while Qwen demonstrated superior legal reasoning capabilities (**71.55% accuracy**), Llama proved more reliable for specific source attribution. Our "Oracle RAG" experiments further revealed that the primary performance bottleneck lies in the model’s reasoning logic rather than context retrieval. Consequently, while the full fine-tuning pipeline remains a target for future optimization, we conclude that a RAG architecture is immediately essential to ensure the source-grounded transparency required for high-stakes legal applications.

V. DISCUSSION

Our experimental analysis heavily prioritizes the evaluation of the Retrieval-Augmented Generation (RAG) component over the full fine-tuning pipeline. This focus was driven by the significant disparity in iteration cycles between the two approaches. While the RAG framework allowed for rapid, training-free evaluation of hypothesis changes, the Continual Pre-Training (CPT) and Instruction Fine-Tuning (IFT) stages required 3 to 6 hours per training run, exclusive of the extensive time required for synthetic dataset generation. Furthermore, a significant portion of the project timeline was dedicated to diagnosing early performance degradation, which we traced to insufficient domain specificity in the initial datasets. By the time we established a robust methodology for CPT and IFT, the project constraints precluded the high-frequency iterative testing required to fully optimize these stages. Consequently, we prioritized the rigorous validation of the RAG mechanism, which offered immediate, measurable insights into source attribution accuracy, and we designate the validation of the complete end-to-end pipeline (Base Model → CPT → IFT → RAG) as the primary objective for future development.

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ETHICAL RISKS

The primary ethical risk identified in this project is the possibility that the language model produces incorrect or misleading answers to questions related to IHL. This risk primarily affects civilian populations, as erroneous interpretations of IHL could contribute to the normalization, justification, or misunderstanding of unlawful conduct during armed conflict. A user who relies on and trusts the model’s outputs could potentially justify harmful actions by citing the model as an authoritative source.

A particularly concerning aspect of this risk is that the model can generate responses that are not only incorrect, but also confident, coherent, and well-structured. In several observed cases, the model’s answers directly contradict fundamental IHL principles. For example, in Figure 4, the model suggests that targeting a military objective is permissible even when excessive civilian casualties are expected. In another case (Figure 5), the model fails to identify a false statement despite citing a seemingly relevant legal rule. These failures often arise from subtle logical errors, such as misinterpreting absolute terms like “always” or “never,” which are critical in legal reasoning.

The severity of this risk is high. In the worst case, a user who fully trusts the model despite its failures could use its output to justify actions that violate IHL, potentially contributing to serious harm to civilians. Although the likelihood of such misuse is currently low—given that the model is not publicly deployed—the likelihood of incorrect answers remains significant, with an observed error rate of approximately 30%. But many errors involve partial inaccuracies rather than complete reversals of meaning.

We evaluated this risk through systematic benchmarking on the ICRC MCQs. We evaluated answer accuracy and source attribution accuracy. Manual inspection of incorrect outputs allowed us to identify recurring reasoning errors and instances of source hallucination.

This risk was explicitly taken into account in the project design. We introduced a RAG to ground the model’s outputs in authoritative IHL rules and refined prompting strategies to enforce stricter logical constraints. However, the risk could not be fully eliminated. Even when provided with the correct legal source and carefully designed prompts, the model may still misapply the rule. This highlights a fundamental limitation of current language models in high-stakes legal reasoning, where a single misinterpreted word can invert the meaning of an answer.

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APPENDIX A. PROMPT TEMPLATES

A. Classical Evaluation Prompt

Prompt Used for IHL Question Answering

```

1 You are a legal expert taking a
2 multiple-choice exam.
3
4 Question: {question}
5
6 A) {options['A']}
7 B) {options['B']}
8 C) {options['C']}
9 D) {options['D']}
10
11 Think quickly step by step and Select
12 the best option or options and reply
13 with the final answer inside
14 <answer></answer> tags, here are few
15 examples:
16
17 <answer>A</answer>,
18 <answer>C</answer>,
19 <answer>B,C,D</answer>,
20 <answer>A,B,C,D</answer>

```

B. RAG Evaluation Prompt

Prompt Used for IHL Question Answering using RAG

```

1 You are a legal expert in International
2 Humanitarian Law.
3 Answer the MCQs based on the provided
4 context.
5
6 ### CONTEXT:
7 {context}
8
9 ### QUESTION:
10 {question}
11
12 ### OPTIONS:
13 A) {options['A']}
14 B) {options['B']}
15 C) {options['C']}
16 D) {options['D']}
17
18 ### INSTRUCTIONS:
19 Think quickly step by step and then
20 provide the MCQ answer inside
21 <answer></answer> tags in the format:
22
23 <answer>A</answer>
24 or
25 <answer>A,C</answer>
26 or
27 <answer>A,C,B</answer>
28 or
29 <answer>A,C,D,B</answer>
30
31 There can be other permutations of the
32 examples provided for the correct answers
33
34 Also provide the IHL source that

```

```

35 correspond the best inside
36 <source></source>.
37
38 Example: <source>Rule 123</source>
39 (could be multiple sources)
40
41 ### ANSWER:

```

APPENDIX B. OUTPUT EXAMPLES

1st Example of Incorrect Model Output

```

1 "question_number": 197,
2 "question": "Under IHL, it is always permissible
   to:",
3 "options": {
4   "A": "Target an adversary's military
       objectives, even if the attack is
       expected
       to cause excessive incidental civilian
       casualties.",
5   "B": "Conduct surprise attacks on civilian
       infrastructure if they offer indirect
       support to the enemy's military efforts.",
6   "C": "Attack dual-use objects if their
       neutralization offers a definite military
       advantage and incidental civilian harm is not
       excessive.",
7   "D": "Target civilian populations, as long as
       done in self-defense against an
       occupying power."},
8 "predicted":
9   "A"
10 "ground_truth":
11   "C"
12 <answer>A</answer>
13 Explanation: According to Rule 14, launching an
14 attack that may be expected to cause
15 incidental loss of civilian life, injury to
16 civilians, damage to civilian objects, or
17 a combination thereof, which would be excessive in
18 relation to the concrete and direct
19 military advantage anticipated, is prohibited.
20 This means that it is not always
21 permissible to target an adversary's military
22 objectives if the attack is expected to
   cause excessive incidental civilian casualties."

```

Figure 4. Illustrative failure case showing that the language model can produce a confident but legally incorrect answer. Although the model correctly cites Rule 14, it selects an option that directly contradicts the proportionality requirement stated in the rule.

2nd Example of Incorrect Model Output

```

1 "question_number": 195,
2 "question": "Which of the following statements is
   false regarding the principle of
   proportionality in IHL?",
3 "options": {
4   "A": "A military operation must be canceled if
       the anticipated incidental harm
       to civilians clearly exceeds the expected
       military advantage.",
5   "B": "The principle of proportionality
       requires a balancing between civilian
       harm and military advantage, considering only
       direct and immediate effects

```



```

9   on civilian populations.",
10  "C": "The attacking party has a duty to take
      feasible precautions to avoid or
11  minimize incidental harm to civilians.",
12  "D": "Indirect consequences, such as economic
      disruptions, are relevant when
13  assessing proportionality under IHL."
14 },
15 "predicted": [
16   "A"
17 ],
18 "ground_truth": [
19   "B"
20 ],
21 "raw_output": "<source>Rule 18</source>
22
23 <answer>A</answer>
24
25 Explanation: According to Rule 18, each party to
26 the conflict must do everything
27 feasible to assess whether the attack may be
28 expected to cause incidental loss of
29 civilian life, injury to civilians, damage to
30 civilian objects, or a combination
31 thereof, which would be excessive in relation to
   the concrete and direct military
   advantage anticipated. However, the statement \"A
   military operation must be
   canceled if the anticipated incidental harm to
   civilians clearly exceeds the
   expected military advantage\" is not explicitly
   mentioned in the rules."

```

Figure 5. Illustrative failure case showing confident but incorrect legal reasoning. Although the model correctly cites Rule 18, it selects an answer that is compatible with proportionality obligations while failing to identify the false restriction introduced in option B.

Example of Confident but Hallucinated Legal Reasoning

```

1 "question":
2 "Can a party to an armed conflict punish a person
3 for disobeying an order to perform
4 an act that would be contrary to medical ethics?"
5
6 "model_raw_response":
7 "To answer this question, we need to refer to the
8 relevant International Humanitarian
9 Law (IHL) rules. The correct answer is based on
10 the principle of medical ethics and the
11 protection of medical personnel.
12
13 The correct answer is:
14
15 <answer>B</answer>
16
17 This answer is based on the following IHL rule:
18
19 <sources>Rule 112</sources>
20
21 This rule states that 'Persons engaged in medical
22 activities may not be compelled to
23 perform acts or to carry out work contrary to the
24 rules of medical ethics or to other
25 rules designed for the benefit of the wounded and
26 sick.'
27
28 The other options are incorrect because:
29 A) Punishment for refusing unethical medical acts
30 is prohibited.
31 C) Medical experimentation concerns a different
32 prohibition.

```

```

25 D) Performing medical duties is protected under
    IHL."
26
27 "ground_truth":
28 {
29   "correct_answers": ["B", "C", "D"],
30   "correct_source": "Rule 26"
31 }
32
33 "error_type":
34 - Hallucinated legal source (Rule 112 does not
    apply)
35 - Partial answer (missed correct options C and D)

```

Figure 6. Failure case illustrating confident but incorrect legal grounding. The model provides a fluent and authoritative justification while hallucinating an inapplicable legal source (Rule 112) and omitting two correct answer options. This example highlights the risk of ungrounded legal reasoning in the absence of strict source verification.

APPENDIX C. DATASETS SAMPLES

Example of Sample in the *charlotte_scraps/output.jsonl* dataset

```

1 {
2   "text": "IN A CRISIS THEY CARE. HONOURING
            EXCEPTIONAL NURSES AND NURSING AIDES.
            FLORENCE NIGHTINGALE MEDAL. 48th Award - 12
            May 2021. CIRCULAR No. 635. Geneva, 1
            September 2020. Florence Nightingale Medal:
            Call for nominations. To the secretaries-
            general and presidents of National
            Societies.
3
4   Dear Sir/Madam, Nurses and nursing aides perform
            extraordinary acts to help the injured and
            sick in times of armed conflict, in the
            wake of natural or man-made disasters, and
            during pandemics. In recognition of this,
            the Florence Nightingale Medal will be
            awarded on 12 May 2021 to candidates who
            have shown exceptional courage and devotion
            to the wounded, sick or disabled, or to
            civilian victims in such crises.
5
6   It will also be awarded in recognition of
            exemplary services or a creative and
            pioneering spirit in the areas of public
            health or nursing education. We believe
            that it is important to celebrate the
            contribution that nurses and nursing aides
            make to the work of the International Red
            Cross and Red Crescent Movement and to
            humanitarian work in general. By ensuring
            that vulnerable people receive medical care
            in some of the most difficult
            circumstances, nurses and nursing aides
            embody the spirit of humanity, one of our
            Fundamental Principles.
7
8   It is with great pleasure that I invite the
            secretaries-general and presidents of
            National Societies to nominate candidates
            who have distinguished themselves in this
            way. Terms for nominating candidates:
            Candidates for the Florence Nightingale
            Medal must be nominated in accordance with
            the regulations governing the awarding of
            the medal. The regulations and guidance for
            National Societies on how to identify
            candidates and submit applications are

```

attached to this circular.

The significance of the Florence Nightingale Medal can only be maintained if the secretaries-general and presidents of National Societies take the greatest of care in selecting candidates. Applications that have not been signed by a president or secretary-general cannot be accepted. The specific grounds for the nomination should also be stated, in particular the circumstances in which the candidate has shown exceptional courage and devotion or in which he or she has displayed exemplary moral and professional conduct.

Moreover, a description should be given of the candidate's current and past responsibilities in order to provide a full picture of his or her qualities. Please note that if the candidate has retired or is deceased, no more than two years may have elapsed between the end of his or her activities and the date on which the nomination is submitted. In addition, the medal cannot be awarded posthumously unless the candidate died while carrying out his or her duties.

Consultation with national associations of nurses: National Societies are encouraged to ask their national associations of nurses to recommend candidates for consideration. Although applications can be made only by National Societies, candidates from outside the Movement are welcome.

Nomination process: The secretaries-general and presidents of National Societies are invited to submit their nominations to the International Committee of the Red Cross (ICRC), for the attention of Mr Emmanuel Seite, secretary of the Florence Nightingale Medal Commission, at the following address: eseite@icrc.org. If they wish, they may put forward more than one candidate. The deadline is 1 February 2021.

Attached is the nomination form, which can be completed in Arabic, English, French or Spanish. Nominations not received by the stated deadline may be resubmitted for consideration in 2023. We would like to emphasize that the Florence Nightingale Medal is not intended to crown a career. We are sure you will make good use of this opportunity to recognize and honour individuals eligible for this distinction and, in so doing, inspire your colleagues to continue their important work. If you have any questions or would like assistance in preparing the nominations, please contact the ICRC's cooperation or health delegates in your country. Yours sincerely, Peter Maurer, President of the ICRC. Jacques Chapuis, President of the Florence Nightingale Medal Commission."

Example of Sample in the *legitron-mixture-instruction-tuning.jsonl* dataset

```
1 {
2
3   "conversations": [
4
5     {"from": "user", "value": "is there a risk of my
6      credit card information being leaked
7      or misused?\nTerm: Call Recording and Monitoring
8      ."},
9
10    {"from": "assistant", "value": "### Risk of Credit
11      Card Information Leakage or Misuse
12      \n\nThe term **Call Recording and Monitoring** is
13      unrelated to the risk of credit
14      card information leakage or misuse. This term
15      typically refers to the practice of
16      recording and monitoring phone conversations for
17      quality control, security, or
18      compliance purposes.\n\n### What You Should Know\n
19      \n*   Call recording and monitoring
20      systems usually don't have access to sensitive
21      information like credit card numbers.\n
22      *   In most cases, these systems are designed to
23      capture audio or voice data, not
24      financial information.\n*   If you're concerned
25      about the security of your credit card
26      information, it's best to reach out to your bank
27      or credit card issuer directly."}]
28 }
```

Example of Sample in the *legitron-mixture-text.jsonl* dataset

```
1 {"text": "site: media | arena: nfl | pageType:
2   stories | section: | slug: brian-urlacher-ray-
3   -lewis-is-the-best-middle-linebacker-of-all-
4   time | sport: football | route:
5   article_single.us | 6-keys: media/spln/nfl/
6   reg/free/stories
7
8   Brian Urlacher: Ray Lewis is the best middle
9   linebacker of all time. Chicago Bears
10  linebacker Brian Urlacher, who announced
11  his retirement on Wednesday, said that he
12  considers former Ravens linebacker Ray
13  Lewis to be the best middle linebacker of
14  all time.
15
16  By John Breech. May 23, 2013 at 4:04 pm ET - 1
17  min read. Brian Urlacher believes that Ray
18  Lewis is the best middle linebacker in NFL
19  history. (USATSI). If you ever find
20  yourself in a debate with someone about who
21  the better middle linebacker was between
22  Brian Urlacher and Ray Lewis, pick Ray
23  Lewis. That's who Urlacher would pick.
24  Urlacher, who announced his retirement on
25  Wednesday, thinks Lewis is not only better
26  than him but better than every middle
27  linebacker who has ever played the game.
28
29  As long as neither player comes out of
30  retirement, both men would become eligible
31  for the Hall of Fame in 2018, \"If I'm
32  lucky enough to go in with Ray, I'll say
33  this about Ray: I think I'm pretty good;
34  Ray's the best of all time,\" Urlacher said
```


on the NFL Network's Total Access show. \"His numbers, guys who played with him, you watch him, he did it the right way, he played hard. He's the best middle linebacker of all time.\"

Those are strong comments, especially coming from Urlacher, whose Bears have produced some of the best middle linebackers in NFL history, including Mike Singletary and Dick Butkus. \"He was great,\" Urlacher said of Lewis. \"I'm up there, I think, but not quite. The longevity he had was crazy, 17 years.\"

Urlacher played 13 seasons in the NFL, a number that might have been 14 if the Bears hadn't fired Lovie Smith. Urlacher thinks he would have played at least one more season if Smith was still coaching in Chicago. \"I do believe that. Yes,\" Urlacher said. \"And I think my desire would be a lot more, as well. Nothing against the new coaching staff. When you start playing for a guy like Lovie, you want to keep playing.\"

Urlacher finished his career with eight Pro Bowls along with a Defensive Rookie of the Year award (2000) and a Defensive Player of the Year award (2005). Lewis trumped that output with 13 Pro Bowl appearances, two Defensive Player of the Year awards (2000, 2003), two Super Bowl wins and a Super Bowl MVP award in 2000. John Breech 1 min read

the ICRC as an indiscriminate attack . Such attacks are serious violations of IHL as they fail to distinguish between civilian and military targets .\n4. Conclusion: The attack on Najaf , resulting in civilian casualties, is a violation of international humanitarian law as per the ICRC's statement.\n</think>\n\nYes, the attack on Najaf is a violation of international humanitarian law."

```
14     }
15   },
16   "source_doc_id": 6562
17 }
```

APPENDIX D. QUESTION DISTRIBUTIONS

The IHL benchmark specifically includes questions sourced from:

- 2003_ICRC_challenges_report (24 questions),
- 2007_ICRC_challenges_report (52 questions),
- 2011_ICRC_challenges_report (29 questions),
- 2015_ICRC_challenges_report (26 questions),
- 2019_ICRC_challenges_report (37 questions),
- 2024_ICRC_challenges_report (33 questions),
- Rules 1--161 (262 questions).
- unknown (8 questions).

Example of Sample in the *ift_vLLM_charlotte_gwen_full_v2.json* (IRAC generated) dataset

```
1 {
2   "conversations": [
3     {
4       "from": "system",
5       "value": "You are an intelligent legal
6                 assistant specialized in
7                 International Humanitarian Law. You
8                 assume the role of a jurist utilizing
9                 the IRAC method."
10    },
11    {
12      "from": "human",
13      "value": "Is the attack on the holy city
14                of Najaf, resulting in civilian
15                casualties, a violation of
16                international humanitarian law
17                according to the ICRC's statement?"
18    },
19    {
20      "from": "gpt",
21      "value": "<think>\nStep-by-step reasoning
22                using IRAC format:\n1. Issue: Whether
23                the attack on Najaf that caused
24                civilian casualties violates
25                international humanitarian law.\n2.
26                Rule: (Cite the text) The ICRC states
27                that 'indiscriminate attacks in Iraq
28                , which negate the basic principles
29                of humanity and are serious
30                violations of international
31                humanitarian law.'\n3. Application:
32                The attack on Najaf led to civilian
33                casualties, which is characterized by
```